

CCNA

VLAN and INTER-VLAN Routing TASK – 5

1. On a Cisco Packet Tracer or GNS3 simulation, create three VLANs named Insta, Zomato, and Flipkart on a single switch, and assign at least one PC to each VLAN.

Ans :

- enable
- configure terminal

- vlan 10
- name Insta
- exit

- vlan 20
- name Zomato
- exit

- vlan 30
- name Flipkart
- exit

- interface fastEthernet 0/1
- switchport mode access
- switchport access vlan 10
- exit

- interface fastEthernet 0/2
- switchport mode access

- switchport access vlan 20
- exit

- interface fastEthernet 0/3
- switchport mode access
- switchport access vlan 30
- exit

- end
- show vlan brief

2. Configure a Router-on-a-Stick setup by creating sub-interfaces (e.g., GigabitEthernet0/0.10, .20, .30) on a simulated router, each with an IP address for the corresponding VLANs Insta, Zomato, and Flipkart.

Ans :

- enable
- configure terminal
- interface gigabitEthernet 0/0
- no shutdown
- exit
- interface gigabitEthernet 0/0.10
- encapsulation dot1Q 10
- ip address 192.168.10.1 255.255.255.0
- exit
- interface gigabitEthernet 0/0.20
- encapsulation dot1Q 20
- ip address 192.168.20.1 255.255.255.0
- exit

- interface gigabitEthernet 0/0.30
- encapsulation dot1Q 30
- ip address 192.168.30.1 255.255.255.0
- exit

- end
- show ip interface brief

- enable
- configure terminal

- interface fastEthernet 0/24
- switchport mode trunk
- no shutdown
- exit

- end
- show interfaces trunk

**3. Assign the switch port connected to the router as a trunk port, and verify trunking is active using the show interfaces trunk command.

Hint: Use the switchport mode trunk command on the relevant interface.**

Ans :

- enable
- configure terminal
- interface fastEthernet 0/24
- switchport mode trunk
- no shutdown
- exit

- end
- show interfaces trunk

4. Test inter-VLAN connectivity by pinging from a PC in the Insta VLAN to a PC in the Zomato VLAN, and explain why the ping succeeds or fails based on your configuration.

Ans :

Insta PC

192.168.10.10

|

VLAN 10

|

Switch

|

Trunk Fa0/24

|

Router G0/0.10

192.168.10.1

|

Routing

|

Router G0/0.20

192.168.20.1

|

VLAN 20

|

Zomato PC

192.168.20.10

5. Refactor your Router-on-a-Stick configuration to add a new VLAN called 'Spotify' without disrupting existing VLAN communication, and document the exact commands you used for both the switch and the router.

Ans :

- enable
- configure terminal

- vlan 40
- name Spotify
- exit

- interface fastEthernet 0/4
- switchport mode access
- switchport access vlan 40
- no shutdown
- exit

- interface fastEthernet 0/24
- switchport mode trunk
- no shutdown
- exit

- end
- show vlan brief
- show interfaces trunk

- enable
- configure terminal
- interface gigabitEthernet 0/0
- no shutdown
- exit
- interface gigabitEthernet 0/0.40
- encapsulation dot1Q 40
- ip address 192.168.40.1 255.255.255.0
- no shutdown
- exit
- end
- show ip interface brief